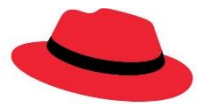


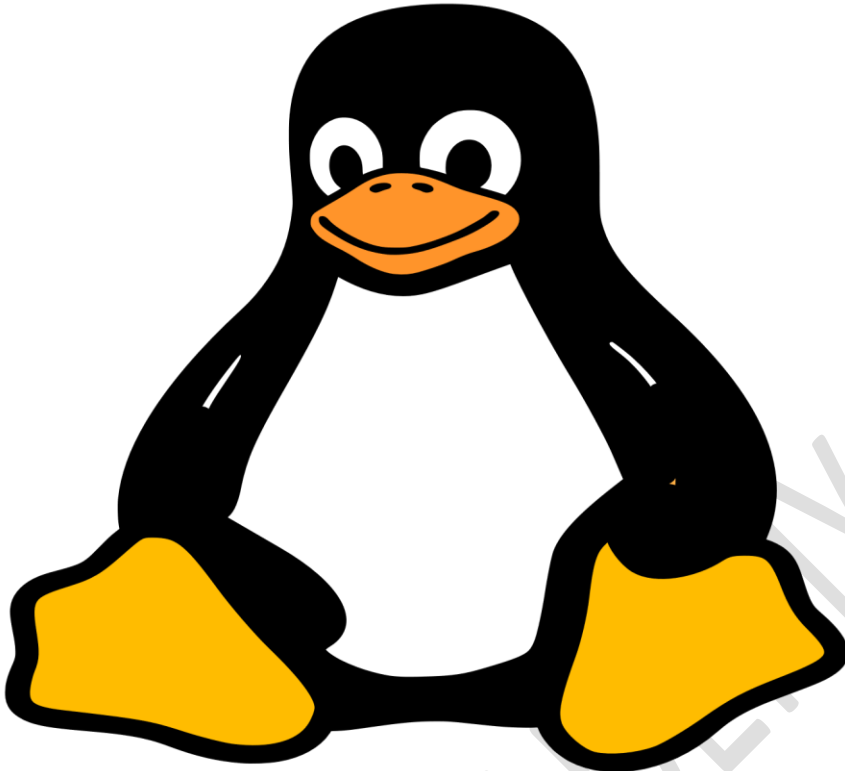


Red Hat

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Red Hat



Red Hat Linux is a popular and influential distribution of the Linux operating system, developed and maintained by Red Hat, Inc., a subsidiary of IBM. Known for its robust performance, enterprise-level features, and extensive support, Red Hat Linux has established itself as a key player in the enterprise Linux market.

Key Features

- Enterprise-Grade Stability:** Red Hat Linux, particularly its flagship distribution Red Hat Enterprise Linux (RHEL), is renowned for its stability and reliability, making it a preferred choice for mission-critical environments.
- Support and Services:** Red Hat offers extensive support services, including regular updates, security patches, and dedicated support channels, which are critical for enterprise users.
- Security:** With a strong focus on security, Red Hat Linux incorporates various security features such as SELinux (Security-Enhanced Linux), regular vulnerability assessments, and compliance with industry standards.
- Ecosystem and Integration:** Red Hat provides a comprehensive ecosystem that includes middleware, cloud solutions, container orchestration with OpenShift, and integration with major cloud providers like AWS, Azure, and Google Cloud.
- Open Source and Community:** While Red Hat Linux itself is a commercial product, it is built on open-source principles. CentOS Stream and Fedora serve as upstream projects for RHEL, fostering community contributions and innovation.
- Subscription Model:** Red Hat uses a subscription-based model that provides access to software updates, security patches, and professional support, ensuring users have the latest features and fixes.

Versions and Variants

- **Red Hat Enterprise Linux (RHEL):** The main product line designed for enterprises, offering long-term support and stability.
- **Fedora:** A community-driven upstream project that serves as a testing ground for new features and technologies that may later be included in RHEL.
- **CentOS Stream:** Positioned between Fedora and RHEL, CentOS Stream provides a rolling-release model that continuously updates with minor changes, acting as a preview of the next RHEL version.

Use Cases

- **Servers and Data Centers:** Due to its robustness and scalability, RHEL is widely used in servers and data centers.
- **Cloud Computing:** Integration with major cloud platforms and support for containerization technologies like Docker and Kubernetes make Red Hat Linux a strong contender in cloud computing environments.
- **Development and Testing:** Developers use Fedora and CentOS Stream for creating and testing new applications, leveraging the latest technologies and features.

Getting Started

To get started with Red Hat Linux:

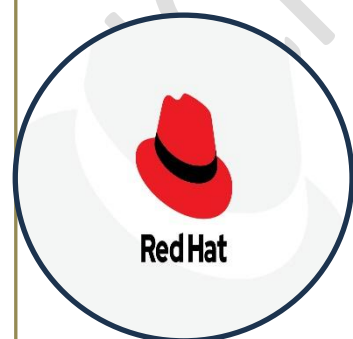
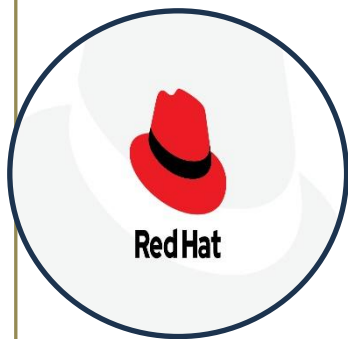
1. **Download and Installation:** Obtain the software from Red Hat's official website or through a subscription. Fedora and CentOS Stream can be downloaded freely.
2. **Subscription and Support:** For RHEL, purchase a subscription to access updates and support.
3. **Community and Resources:** Engage with the community through forums, user groups, and official documentation to leverage the collective knowledge and support.

Conclusion

Red Hat Linux's combination of enterprise-grade features, strong security, extensive support, and commitment to open source makes it a leading choice for organizations seeking a reliable and scalable operating system for their infrastructure.



The Red Hat Certified System Administrator (RHCSA) is a certification provided by Red Hat, Inc. that validates the skills and knowledge necessary for a system administrator to effectively manage Red Hat Enterprise Linux (RHEL) systems. The RHCSA certification is highly regarded in the IT industry and serves as a foundational credential for those pursuing careers in Linux system administration.



Red Hat, Inc., known for its development and distribution of Red Hat Linux, has a rich history that reflects its growth from a small startup to a major player in the enterprise software market. Here's an overview of its history:

Early Years (1993-1999)

- **1993:** Marc Ewing created the original Red Hat Linux distribution and released it in October. The name "Red Hat" comes from Ewing's experience with his grandfather's red Cornell University lacrosse hat, which he wore during his time at Carnegie Mellon University.
- **1994:** Ewing partnered with Bob Young, who had been running a catalog business called ACC Corporation that sold Linux and UNIX software accessories. Together, they formed Red Hat Software and combined their efforts to promote and expand Red Hat Linux.
- **1995:** The company released Red Hat Linux 2.0, which became notable for its use of the RPM Package Manager (RPM), allowing for easier software management and installation.

Growth and IPO (1999-2009)

- **1999:** Red Hat went public, making one of the most successful initial public offerings (IPO) of the dot-com era. This brought significant attention and capital to the company.
- **2000:** Red Hat expanded its product offerings and services, emphasizing enterprise solutions and support.
- **2003:** Red Hat introduced Red Hat Enterprise Linux (RHEL), shifting its focus from the consumer market to the enterprise sector. This move aimed to provide a stable, secure, and supported Linux distribution tailored for business needs.

Expansion and Acquisitions (2009-2018)

- **2006:** Red Hat acquired JBoss, an open-source middleware provider, enhancing its capabilities in enterprise software.
- **2011:** The company acquired Gluster, a software-defined storage company, further expanding its portfolio.
- **2014:** Red Hat acquired eNovance, a provider of open-source cloud computing services, strengthening its position in the cloud market.
- **2015:** The acquisition of Ansible, an IT automation company, marked another significant step in expanding Red Hat's cloud and DevOps capabilities.

Acquisition by IBM (2018-Present)

- **2018:** IBM announced its intent to acquire Red Hat for approximately \$34 billion, one of the largest software company acquisitions in history. This move aimed to bolster IBM's hybrid cloud strategy.
- **2019:** The acquisition was completed in July. Red Hat continues to operate as an independent unit within IBM, with a focus on open-source solutions, hybrid cloud infrastructure, and enterprise-grade software.

Impact and Legacy

- **Open Source Leadership:** Throughout its history, Red Hat has been a strong advocate for open-source software, contributing to and supporting various open-source projects and communities.
- **Enterprise Solutions:** Red Hat's transition to enterprise solutions with RHEL and subsequent acquisitions positioned the company as a leader in enterprise IT infrastructure, cloud computing, and automation.
- **Global Presence:** Red Hat has established a global presence, with offices and partners worldwide, serving a diverse range of industries and sectors.

Conclusion

Red Hat's journey from a small startup to an influential enterprise software company highlights its adaptability, commitment to open-source principles, and strategic expansion. Its acquisition by IBM underscores its value and the critical role it plays in the broader IT landscape.

Certification Objectives

The RHCSA certification focuses on essential system administration tasks that are required to manage RHEL systems, including:

1. Understanding and Using Essential Tools:

- Accessing the command line.
- Using basic file manipulation tools.
- Understanding file permissions and security.

2. Operating Running Systems:

- Booting, rebooting, and shutting down systems.
- Identifying processes, starting, and stopping virtual machines.
- Managing system services and configuration.

3. Configuring Local Storage:

- Partitioning and formatting disk devices.
- Creating and managing logical volumes.
- Mounting and unmounting filesystems.

4. Creating and Configuring File Systems:

- Managing file permissions and ownership.
- Creating and configuring encrypted file systems.

5. Deploying, Configuring, and Maintaining Systems:

- Installing and updating software packages.
- Configuring local storage using partitions and logical volumes.
- Managing users and groups.
- Managing security, including basic firewall and SELinux configuration.

6. Managing Basic Networking:

- Configuring network interfaces.
- Understanding and using network utilities.
- Configuring basic firewall settings.

Exam Details

- **Exam Code:** EX200
- **Format:** Performance-based practical exam.
- **Duration:** 2.5 hours.
- **Passing Score:** Typically, a score of 210 out of 300 is required to pass, though this can vary slightly.

Preparation

- **Official Training:** Red Hat offers official training courses such as Red Hat System Administration I (RH124) and Red Hat System Administration II (RH134), which cover the necessary skills and knowledge for the RHCSA exam.
- **Practice:** Hands-on practice is crucial. Set up a lab environment to practice system administration tasks.
- **Study Guides and Books:** Various study guides and books are available to help prepare for the exam, including the official Red Hat RHCSA/RHCE study guides.
- **Online Resources:** Utilize online resources, forums, and communities for additional support and information.

Benefits

- **Career Advancement:** RHCSA certification can enhance job prospects and career advancement opportunities in the IT field.
- **Validation of Skills:** It provides formal recognition of your skills in managing RHEL systems.
- **Foundation for Advanced Certifications:** RHCSA is a prerequisite for the Red Hat Certified Engineer (RHCE) certification and other advanced Red Hat certifications.

RH124 - Red Hat System Administration – I

1. Getting started with Red Hat Enterprise Linux
2. Access the command line
3. Manage files from the command line
4. Get help in Red Hat Enterprise Linux
5. Create, view & edit text files
6. Manage local users & groups
7. Control access to files
8. Monitor & manage Linux processes
9. Control services & daemons
10. Configure & secure SSH
11. Analyze & store logs
12. Manage networking
13. Archive & transfer files
14. Install & update software
15. Access Linux file systems
16. Analyze servers & get support
17. Comprehensive review

RH134 - Red Hat System Administration - II

1. Improving command line productivity using shell scripts
2. Schedule future tasks
3. Tune system performance
4. Manage SELinux security
5. Maintain basic storage
6. Manage storage stack
7. Access network-attached storage
8. Control the boot process
9. Manage network security
10. Install Red Hat Enterprise Linux
11. Run Containers
12. Comprehensive Review

• 1. Getting started with Red Hat Enterprise Linux

1. Understanding RHEL

RHEL is a robust and stable Linux distribution designed for enterprise environments. It's known for its security, scalability, and support from Red Hat, Inc.

2. Obtaining RHEL

- **Subscription:** To get RHEL, you need a subscription. Red Hat offers various subscription plans, including a free developer subscription which gives you access to RHEL for individual use.
- **Download:** Once you have a subscription, download the RHEL ISO from the Red Hat Customer Portal.

3. Preparing for Installation

- **Hardware Requirements:** Ensure your hardware meets the minimum requirements (2GB RAM, 20GB disk space, etc.).
- **Bootable Media:** Create a bootable USB drive or DVD with the RHEL ISO using tools like Rufus (Windows) or `dd` (Linux).

4. Installing RHEL

1. **Boot from the Installation Media:** Insert the bootable USB/DVD and boot your machine.
2. **Installation Process:**
 - Select "Install Red Hat Enterprise Linux" from the boot menu.
 - Choose your language and keyboard layout.
 - Set up installation source and software selection. You can choose a minimal install or include additional environments like a server with GUI.
 - Configure your disk partitions. You can use automatic partitioning or customize it according to your needs.
 - Set up your network and hostname.
 - Create a root password and, optionally, a user account.
3. **Complete Installation:** Begin the installation and wait for it to complete. After installation, the system will prompt you to reboot.

5. Post-Installation Configuration

- **First Boot:** After rebooting, log in with the root or user account you created.
- **Register with Red Hat:** Use `subscription-manager register` to register your system with Red Hat and enable repositories.
- **System Update:** Run `yum update` or `dnf update` to ensure your system is up to date.

6. Basic System Administration

- **Package Management:** Use `yum` or `dnf` for package management.
 - `dnf install package_name`: Install a package.
 - `dnf remove package_name`: Remove a package.
 - `dnf update`: Update all installed packages.
- **User Management:**
 - `useradd username`: Add a new user.
 - `passwd username`: Set a password for the user.
 - `usermod -aG group username`: Add a user to a group.
- **Service Management:**
 - `systemctl start service_name`: Start a service.
 - `systemctl stop service_name`: Stop a service.
 - `systemctl enable service_name`: Enable a service to start at boot.
 - `systemctl status service_name`: Check the status of a service.

7. Learning Resources

- **Official Documentation:** The Red Hat Customer Portal offers extensive documentation and guides.
- **Training and Certification:** Red Hat provides training programs and certifications like RHCSA and RHCE.
- **Community and Support:** Engage with the community through forums and mailing lists. Utilize Red Hat support if you have a subscription.

8. Advanced Topics

- **SELinux:** Learn about Security-Enhanced Linux for enforcing security policies.
- **Firewalld:** Manage firewall rules with `firewalld`.
- **Virtualization:** Use KVM for creating and managing virtual machines.
- **Containers:** Explore containerization using Podman, which is included in RHEL.

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• 2. Access the command line

1. Accessing the Command Line During Installation

- When installing RHEL, you will interact with the command line during the setup process if you choose to perform custom partitioning or other advanced installation steps.

2. Post-Installation Access

After the installation is complete, you can access the command line in several ways:

A. Accessing via Graphical User Interface (GUI)

1. Terminal Application:

- If you installed a GUI environment (like GNOME), you can open the terminal application from the desktop.
- Look for the terminal icon in the application menu or press **Ctrl + Alt + T** to open it.

B. Accessing via Virtual Consoles

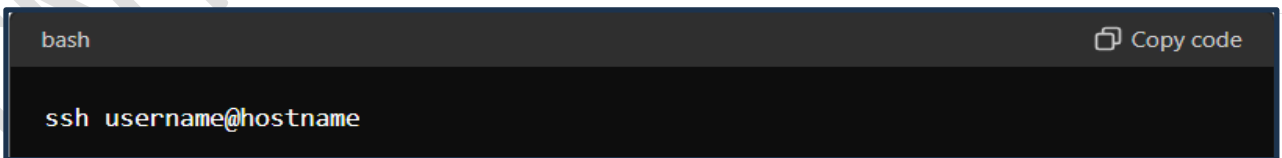
1. Virtual Consoles (TTY):

- RHEL provides multiple virtual consoles that you can access using keyboard shortcuts.
- Press **Ctrl + Alt + F2** to switch to the first virtual console. This will take you to a text-based login prompt.
- You can switch between different TTYs using **Ctrl + Alt + F3** to **Ctrl + Alt + F6**.
- To return to the GUI (if available), use **Ctrl + Alt + F1** or **Ctrl + Alt + F7**.

C. Accessing Remotely

1. SSH (Secure Shell):

- You can access the command line of a remote RHEL server using SSH.
- From another Linux machine or using an SSH client on Windows (such as PuTTY), connect to the server:

A screenshot of a terminal window with a dark background. The prompt 'bash' is visible in the top left corner. In the top right corner, there is a 'Copy code' button with a copy icon. The command 'ssh username@hostname' is entered on the main line of the terminal.

```
bash ssh username@hostname
```

- Replace **username** with your actual user name and **hostname** with the IP address or domain name of the RHEL server.

3. Basic Command Line Usage

Once you have access to the command line, here are some basic commands to get started:

Navigating the File System

- **pwd**: Print the current working directory.
- **ls**: List directory contents.
- **cd /path/to/directory**: Change the current directory.
- **mkdir directory_name**: Create a new directory.
- **rmdir directory_name**: Remove an empty directory.

Managing Files

- **touch file_name**: Create a new empty file.
- **cp source_file destination_file**: Copy a file.
- **mv source_file destination_file**: Move or rename a file.
- **rm file_name**: Delete a file.
- **cat file_name**: Display the contents of a file.
- **nano file_name** or **vi file_name**: Edit a file using a text editor (**nano** is more user-friendly for beginners, while **vi** is more powerful and widely used).

System Administration

- **sudo command**: Run a command with superuser privileges.
- **shutdown -h now**: Shut down the system immediately.
- **reboot**: Reboot the system.
- **top**: Display real-time system processes and resource usage.
- **df -h**: Display disk space usage.
- **free -m**: Display memory usage.

4. Learning More Commands

- **Manual Pages**: Use the **man** command to read the manual pages for other commands.
 - Example: **man ls** provides detailed information about the **ls** command and its options.
- **Help Option**: Most commands support the **--help** option to display a summary of their usage.
 - Example: **ls --help**.

5. Further Resources

- **Official Documentation**: Red Hat's documentation provides in-depth guides and references.
- **Community Forums and Guides**: Websites like Stack Overflow, Linux Questions, and the Red Hat community forums are excellent places to ask questions and find tutorials.

• 3. Manage files from the command line

Managing files from the command line in Red Hat Enterprise Linux (RHEL) involves using a variety of commands to navigate the file system, create, view, edit, copy, move, and delete files and directories. Here's a detailed guide on how to manage files from the command line:

1. Navigating the File System

- ``pwd``: Print the current working directory.

```
bash
```

[Copy code](#)

```
pwd
```

- ``ls``: List the contents of a directory.

```
bash
```

[Copy code](#)

```
ls
```

```
ls -l    # long format
```

```
ls -a    # include hidden files
```

- ``cd``: Change the current directory.

```
bash
```

[Copy code](#)

```
cd /path/to/directory
```

```
cd ..    # move up one directory
```

```
cd ~     # move to the home directory
```

2. Creating and Viewing Files

- ``touch``: Create a new empty file.

```
bash
```

[Copy code](#)

```
touch filename
```

- ``cat``: Concatenate and display the contents of a file.

```
bash
```

[Copy code](#)

```
cat filename
```

- ``less``: View the contents of a file one page at a time.

```
bash
```

[Copy code](#)

```
less filename
```

- ``head``: Display the first few lines of a file.

```
bash
```

[Copy code](#)

```
head filename
```

```
head -n 10 filename    # display the first 10 lines
```

- ``tail``: Display the last few lines of a file.

```
bash
```

[Copy code](#)

```
tail filename
```

```
tail -n 10 filename    # display the last 10 lines
```

3. Editing Files

- ``nano``: A simple text editor.

```
bash
```

[Copy code](#)

```
nano filename
```

- ``vi`` or ``vim``: A more powerful text editor.

```
bash
```

[Copy code](#)

```
vi filename
```

```
vim filename
```

4. Copying and Moving Files

- ``cp``: Copy files or directories.

```
bash
```

[Copy code](#)

```
cp source_file destination_file
```

```
cp -r source_directory destination_directory # copy directories recursively
```

- ``mv``: Move or rename files or directories.

```
bash
```

[Copy code](#)

```
mv source_file destination_file
```

```
mv source_directory destination_directory
```

5. Deleting Files and Directories

- `rm`: Remove files or directories.

bash

 Copy code

```
rm filename
rm -r directory_name # remove directories recursively
rm -f filename       # force removal without prompt
rm -rf directory_name # force removal of directory recursively
```

- `rmdir`: Remove empty directories.

bash

 Copy code

```
rmdir directory_name
```

6. Creating and Managing Directories

- `mkdir`: Create new directories.

bash

 Copy code

```
mkdir directory_name
mkdir -p /path/to/directory # create parent directories as needed
```

- `rmdir`: Remove empty directories.

bash

 Copy code

```
rmdir directory_name
```

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7. Changing File and Directory Permissions

- `chmod`: Change the permissions of a file or directory.

bash

 Copy code

```
chmod 755 filename      # rwxr-xr-x (owner can read/write/execute; others can read/execute)
chmod 644 filename      # rw-r--r-- (owner can read/write; others can read)
chmod +x filename       # add execute permission to a file
```

- `chown`: Change the ownership of a file or directory.

bash

 Copy code

```
sudo chown user:group filename
sudo chown -R user:group directory_name # change ownership recursively
```

8. Viewing File and Directory Information

- `stat`: Display detailed information about a file or directory.

bash

 Copy code

```
stat filename
```

- `du`: Estimate file space usage.

bash

 Copy code

```
du -h filename          # human-readable format
du -sh directory_name    # summary in human-readable format
```

- `df`: Report file system disk space usage.

bash

 Copy code

```
df -h                  # human-readable format
```

9. Finding Files

- `find`: Search for files in a directory hierarchy.

bash

 Copy code

```
find /path/to/search -name "filename"
```

```
find /path/to/search -type d -name "dirname" # find directories
```

- `locate`: Quickly locate files (requires `mlocate` package and database update).

bash

 Copy code

```
locate filename
```

```
updatedb # update the locate database (run with sudo)
```

• 4. Get help in Red Hat Enterprise Linux

Getting help in Red Hat Enterprise Linux (RHEL) is essential for learning how to use the system effectively and troubleshoot issues. Here are several ways to access help and resources within RHEL:

1. Using the Command Line

The command line in RHEL provides several tools and commands to help you understand how to use other commands and utilities.

A. `man` Command

The `man` (manual) command is the primary way to get detailed information about other commands.

- Usage: ``man [command]``

```
bash
```

[Copy code](#)

```
man ls
```

```
man man    # Manual page for the man command itself
```

- **Navigation:** Use the arrow keys to scroll, ``q`` to quit, and ``/`` followed by a search term to search within the man page.

B. ``--help`` Option

Most commands support the ``--help`` option, which provides a brief summary of how to use the command and its options.

- Usage: ``[command] --help``

```
bash
```

[Copy code](#)

```
ls --help
```

```
cp --help
```

C. ``info`` Command

The ``info`` command provides a more detailed and structured documentation for many commands.

- Usage: ``info [command]``

```
bash
```

[Copy code](#)

```
info ls
```

- **Navigation:** Use the arrow keys to scroll, ``q`` to quit, and ``Tab`` to move between links.

2. System Documentation

RHEL includes various system documentation files stored in the ``/usr/share/doc`` directory.

- **Usage:** Navigate to the directory and list its contents to find documentation.

```
bash
```

[Copy code](#)

```
cd /usr/share/doc
```

```
ls
```

3. Red Hat Customer Portal

The Red Hat Customer Portal is a comprehensive resource for official documentation, knowledge base articles, and more.

- **Access:** Red Hat Customer Portal
- **Features:**
 - **Documentation:** Official guides and manuals for installation, administration, security, and more.
 - **Knowledge Base:** Articles on troubleshooting and best practices.
 - **Forums:** Community and expert discussions.
 - **Support:** Access to support cases and subscription management.

4. Online Community and Forums

Engaging with the Red Hat and broader Linux community can provide valuable insights and support.

- **Red Hat Community:** Participate in discussions and ask questions on the Red Hat Customer Portal Discussion Forums.
- **Stack Exchange:** Sites like [Server Fault](#) and [Unix & Linux Stack Exchange](#) are great for specific questions and community support.
- **LinuxQuestions.org:** A popular forum for all Linux-related questions, including RHEL.

5. Training and Certification

Red Hat offers a range of training programs and certifications to help you master RHEL.

- **Red Hat Training:** Official courses and learning paths available at Red Hat Training.
- **Certification:** Certifications like RHCSA (Red Hat Certified System Administrator) and RHCE (Red Hat Certified Engineer).

6. Books and Online Resources

There are numerous books and online tutorials available for learning RHEL.

- **Books:** Look for titles such as "Red Hat Enterprise Linux 8 Essentials" by Neil Smyth or "Red Hat Enterprise Linux 8 Administration" by Miguel Perez Colino.
- **Online Courses:** Platforms like Udemy, Coursera, and Pluralsight offer courses on RHEL.

• 5. Create, view & edit text files

- Creating, viewing, and editing text files are fundamental tasks you'll perform frequently on Red Hat Enterprise Linux (RHEL). Here's a guide on how to manage text files using the command line:
- **1. Creating Text Files**
- You can create text files using several commands and text editors.

Using the ``touch`` Command

The ``touch`` command creates an empty file if it doesn't already exist.

```
bash
```

[Copy code](#)

```
touch filename.txt
```

Using Redirection Operators

You can create and write to a file using redirection operators.

```
bash
```

[Copy code](#)

```
echo "This is a sample text file." > sample.txt
```

This command creates ``sample.txt`` if it doesn't exist and writes "This is a sample text file." to it.

2. Viewing Text Files

There are various commands to view the contents of text files.

Using the ``cat`` Command

The ``cat`` command displays the entire content of a file.

```
bash
```

[Copy code](#)

```
cat filename.txt
```

Using the ``less`` Command

The ``less`` command allows you to view large files one page at a time.

```
bash
```

[Copy code](#)

```
less filename.txt
```

- Use the arrow keys to navigate.
- Press ``q`` to quit.

Using the `more` Command

The `more` command is similar to `less` but with simpler functionality.

```
bash
```

[Copy code](#)

```
more filename.txt
```

Using the `head` Command

The `head` command displays the first few lines of a file.

```
bash
```

[Copy code](#)

```
head filename.txt
```

```
head -n 10 filename.txt # Display the first 10 lines
```

Using the `tail` Command

The `tail` command displays the last few lines of a file.

```
bash
```

[Copy code](#)

```
tail filename.txt
```

```
tail -n 10 filename.txt # Display the last 10 lines
```

3. Editing Text Files

You can edit text files using text editors available on the command line.

Using `nano`

`nano` is a user-friendly text editor.

```
bash
```

[Copy code](#)

```
nano filename.txt
```

- Basic Commands:
 - `ctrl + O`: Save the file.
 - `ctrl + X`: Exit `nano`.
 - `ctrl + K`: Cut a line.
 - `ctrl + U`: Paste a line.

Using `vi` or `vim`

`vi` (or `vim`) is a more powerful text editor, though it has a steeper learning curve.

bash

 Copy code

```
vi filename.txt
vim filename.txt
```

- Basic Commands:
 - Switch to Insert Mode: Press `i` to start editing.
 - Save and Exit: Press `Esc` to return to command mode, then type `:wq` and press `Enter`.
 - Exit Without Saving: Press `Esc`, then type `:q!` and press `Enter`.

Using `gedit`

If you are working in a graphical environment, `gedit` provides a simple GUI text editor.

bash

 Copy code

```
gedit filename.txt &
```

The ampersand (`&`) runs the editor in the background, allowing you to continue using the terminal.

4. Redirecting and Piping Output

You can use redirection and pipes to create and modify files.

Appending to a File

To append text to an existing file:

bash

 Copy code

```
echo "Additional line of text." >> filename.txt
```

Using Pipes

You can pipe the output of one command to another, for example, using `grep` to search within a file and output the results to another file:

bash

 Copy code

```
grep "search_term" inputfile.txt > outputfile.txt
```

• Manage local users & groups

1. Create a New User

- Command: ``useradd``
- Example:

bash

 Copy code

```
sudo useradd username
```

- Optional: Specify home directory, shell, etc.

bash

 Copy code

```
sudo useradd -m -d /home/username -s /bin/bash username
```

2. Set Password for a User

- Command: ``passwd``
- Example:

bash

 Copy code

```
sudo passwd username
```

3. Modify a User

- Command: ``usermod``
- Example:

bash

 Copy code

```
sudo usermod -l newname oldname # Change username  
sudo usermod -aG groupname username # Add user to group
```

4. Delete a User

- Command: ``userdel``
- Example:

bash

 Copy code

```
sudo userdel username
```

- Optional: Remove home directory

bash

 Copy code

```
sudo userdel -r username
```

Groups Management

1. Create a New Group

- Command: ``groupadd``
- Example:

bash

 Copy code

```
sudo groupadd groupname
```

2. Modify a Group

- Command: ``groupmod``
- Example:

bash

 Copy code

```
sudo groupmod -n newgroupname oldgroupname # Rename group
```

3. Delete a Group

- Command: ``groupdel``
- Example:

bash

 Copy code

```
sudo groupdel groupname
```

Viewing User and Group Information

1. List All Users

- Command:

bash

 Copy code

```
cut -d: -f1 /etc/passwd
```

2. List All Groups

- Command:

bash

 Copy code

```
cut -d: -f1 /etc/group
```



3. View User Information

- Command: ``id``
- Example:

```
bash
```

[Copy code](#)

```
id username
```

Example Workflow

Creating a New User with Specific Group and Home Directory

1. Create a new group:

```
bash
```

[Copy code](#)

```
sudo groupadd developers
```

2. Create a new user with a specific home directory and assign to the group:

```
bash
```

[Copy code](#)

```
sudo useradd -m -d /home/johndoe -s /bin/bash -G developers johndoe
```



3. Set password for the new user:

```
bash
```

[Copy code](#)

```
sudo passwd johndoe
```

4. Verify the user and group:

```
bash
```

[Copy code](#)

```
id johndoe
```

Configuration Files

- `/etc/passwd`: Contains user account information.
- `/etc/shadow`: Contains secure user account information, including encrypted passwords.
- `/etc/group`: Contains group information.
- `/etc/gshadow`: Contains secure group account information.

In commands aur configuration files ke through, aap Linux system mein effectively local users aur groups manage kar sakte hain.

7. Control access to files

Controlling access to files in Linux involves using a combination of permissions, user groups, and special attributes. Here's a guide on how to manage file access effectively:

File Permissions

In Linux, each file and directory has three types of permissions:

- **Read (r)**: Allows viewing the contents of the file.
- **Write (w)**: Allows modifying the contents of the file.
- **Execute (x)**: Allows running the file as a program.

These permissions can be set for three categories of users:

- **Owner (u)**: The user who owns the file.
- **Group (g)**: The group that owns the file.
- **Others (o)**: All other users.

Viewing Permissions

To view the permissions of a file, use the `ls -l` command:

```
bash
```

[Copy code](#)

```
ls -l filename
```

The output looks something like this:

```
csharp
```

[Copy code](#)

```
-rw-r--r-- 1 user group 4096 Jan 1 12:00 filename
```

Here, `-rw-r--r--` indicates:

- `-`: Regular file (could be `d` for directory).
- `rw-`: Read and write permissions for the owner.
- `r--`: Read-only permission for the group.
- `r--`: Read-only permission for others.

Changing Permissions

Permissions can be changed using the `chmod` command. There are two ways to set permissions: symbolic mode and numeric mode.

Symbolic Mode

bash

 Copy code

```
chmod u=rwx,g=rx,o=r filename
```

- ``u=rwx``: Sets the owner's permissions to read, write, and execute.
- ``g=rx``: Sets the group's permissions to read and execute.
- ``o=r``: Sets others' permissions to read.

You can also add or remove specific permissions:

bash

 Copy code

```
chmod g+w filename # Adds write permission to the group  
chmod o-r filename # Removes read permission from others
```

Numeric Mode

Permissions are represented by a three-digit octal number. Each type of permission is assigned a value:

- Read: 4
- Write: 2
- Execute: 1

The sum of these values gives the permission. For example:

- ``7`` (rwx): $4 + 2 + 1$
- ``5`` (r-x): $4 + 0 + 1$
- ``4`` (r--): $4 + 0 + 0$

To set permissions using numeric mode:

```
bash Copy code  
  
chmod 755 filename
```

This sets:

- Owner: 7 (rwx)
- Group: 5 (r-x)
- Others: 5 (r-x)

Changing Ownership

To change the owner or group of a file, use the `chown` command:

```
bash Copy code  
  
chown owner:group filename
```

Special Permissions

Setuid and Setgid

- **Setuid (Set User ID):** When set on an executable file, allows users to run the file with the file owner's privileges.

- **Setgid (Set Group ID):** When set on an executable file, allows users to run the file with the file group's privileges. When set on a directory, new files created within the directory inherit the directory's group.

To set the setuid or setgid:

bash

 Copy code

```
chmod u+s filename # Setuid
```

```
chmod g+s filename # Setgid
```

Sticky Bit

When set on a directory, only the file owner, directory owner, or root can delete or rename files within that directory.

bash

 Copy code

```
chmod +t directory
```

Access Control Lists (ACLs)

ACLs provide a more flexible permission mechanism by allowing specific permissions for users and groups.

To use ACLs:

1. Ensure the filesystem supports ACLs (e.g., ext4, xfs).
2. Install the `acl` package (if not already installed)

```
bash
```

[Copy code](#)

```
sudo apt-get install acl
```

3. Use `setfacl` to set ACLs:

```
bash
```

[Copy code](#)

```
setfacl -m u:username:rw filename # Grants read and write permissions to a user  
setfacl -m g:groupname:r filename # Grants read permissions to a group
```

4. View ACLs with `getfacl`:

```
bash
```

[Copy code](#)

```
getfacl filename
```

Summary

- Use `ls -l` to view permissions.
- Use `chmod` to change permissions (symbolic or numeric mode).
- Use `chown` to change file ownership.
- Use `setuid`, `setgid`, and the sticky bit for special permissions.
- Use ACLs for fine-grained control.

8. Monitor & manage Linux processes

Monitoring and managing processes in Linux is crucial for system administration. Here's a comprehensive guide on how to monitor and manage processes:

Monitoring Processes

1. *ps* Command

The `ps` command provides a snapshot of the current processes. Common options include:

bash

 Copy code

```
ps aux      # Lists all processes with detailed information
```

```
ps -ef      # Alternative format for listing all processes
```

2. *top* Command

The `top` command provides a real-time view of the system's resource usage, including CPU and memory.

bash

 Copy code

```
top         # Launches the top interface
```

- Press `q` to quit.
- Press `h` for help.
- Press `k` to kill a process (you will be prompted for the PID).

3. htop Command

`htop` is an enhanced version of `top`, offering an interactive and colorful interface.

```
bash
```

[Copy code](#)

```
htop      # Launches the htop interface
```

- Use arrow keys to navigate.
- Press `F9` to kill a process.
- Press `q` to quit.

4. vmstat Command

The `vmstat` command reports information about processes, memory, paging, block I/O, traps, and CPU activity.

```
bash
```

[Copy code](#)

```
vmstat 2 5  # Updates every 2 seconds, for a total of 5 times
```

5. pidstat Command

The `pidstat` command provides detailed statistics about CPU usage by individual processes.

```
bash
```

[Copy code](#)

```
pidstat      # Displays CPU usage of processes
```

6. iostat Command

The `iostat` command reports on CPU and I/O statistics.

```
bash
```

[Copy code](#)

```
iostat 2     # Updates every 2 seconds
```

7. sar Command

The `sar` command collects, reports, or saves system activity information.

```
bash
```

[Copy code](#)

```
sar -u 2 5   # CPU usage updated every 2 seconds, 5 times
```

Managing Processes

1. kill Command

The `kill` command sends a signal to a process, typically to terminate it.

bash

 Copy code

```
kill PID          # Sends SIGTERM (15) to PID
kill -9 PID       # Sends SIGKILL (9) to PID, forcing termination
kill -l           # Lists all available signals
```

2. killall Command

The `killall` command kills processes by name.

bash

 Copy code

```
killall process_name # Sends SIGTERM to all processes named process_name
killall -9 process_name # Sends SIGKILL to all processes named process_name
```

3. pkill Command

The `pkill` command sends a signal to processes based on a name or other attributes.

bash

 Copy code

```
pkill process_name # Sends SIGTERM to processes named process_name
pkill -9 process_name # Sends SIGKILL to processes named process_name
```

4. nice and renice Commands

These commands adjust the scheduling priority of a process.

bash

 Copy code

```
nice -n 10 command # Starts command with a niceness of 10 (lower priority)
renice +10 PID      # Changes the niceness of an existing process
```

5. nohup Command

The `nohup` command runs another command immune to hangups, with output to a non-tty.

bash

 Copy code

```
nohup command & # Runs command in the background, immune to hangups
```

6. bg and fg Commands

These commands move jobs to the background and foreground.

```
bash Copy code  
  
command &          # Starts command in the background  
jobs               # Lists current jobs  
bg %1              # Resumes job 1 in the background  
fg %1              # Brings job 1 to the foreground
```

7. systemctl Command

`systemctl` is used to manage systemd services and processes.

```
bash Copy code  
  
systemctl start service    # Starts a service  
systemctl stop service     # Stops a service  
systemctl restart service  # Restarts a service  
systemctl status service   # Checks the status of a service
```

Additional Tips

- **Logging and Monitoring Tools:** Tools like `syslog`, `journalctl`, and monitoring systems like Nagios, Zabbix, and Prometheus can provide comprehensive logging and monitoring solutions.
- **Script Automation:** Use shell scripts to automate repetitive tasks related to process management.
- **Resource Limits:** Configure resource limits using `ulimit` to control the resources available to the shell and its processes.

Understanding Disk Management in Linux

Disk management in Linux involves:

- Viewing disk information
 - Creating/deleting partitions
 - Formatting partitions with filesystems
 - Mounting/unmounting partitions
 - Checking disk usage
 - Managing logical volumes (LVM)
-

2. Viewing Disk Information

Linux provides several commands to see disks and partitions:

1. **lsblk** – Lists all block devices in a tree view

```
lsblk
```

Example output:

NAME	MAJ:MIN	RM	SIZE	RO	TYPE	MOUNTPOINT
sda	8:0	0	100G	0	disk	
├─sda1	8:1	0	50G	0	part	/
├─sda2	8:2	0	20G	0	part	/home
└─sda3	8:3	0	30G	0	part	[SWAP]

2. **fdisk -l** – Lists all disks and partitions

```
sudo fdisk -l
```

3. **df -h** – Shows mounted filesystems with size and usage

```
df -h
```

4. **parted** – For GPT disks and detailed info

```
sudo parted -l
```

3. Partitioning a Disk

Using **fdisk** (for MBR disks)

```
sudo fdisk /dev/sdb
```

Interactive options:

- n → create a new partition
- d → delete a partition
- p → print partition table
- w → write changes

Using parted (for GPT disks)

```
sudo parted /dev/sdb
(parted) mklabel gpt      # create GPT table
(parted) mkpart primary ext4 0% 100% # create partition
(parted) quit
```

4. Formatting Partitions

After creating a partition, format it with a filesystem:

```
sudo mkfs.ext4 /dev/sdb1 # ext4 filesystem
sudo mkfs.xfs /dev/sdb1  # XFS filesystem
sudo mkfs.vfat /dev/sdb1 # FAT32 filesystem
```

5. Mounting and Unmounting

- **Mount a partition:**

```
sudo mount /dev/sdb1 /mnt
```

- **Unmount:**

```
sudo umount /mnt
```

- To mount automatically at boot, edit /etc/fstab.

Example /etc/fstab entry:

```
/dev/sdb1    /data    ext4    defaults    0 2
```

6. Checking Disk Usage

- **df -h** → Disk space of mounted filesystems
 - **du -sh /path/to/folder** → Size of specific folder
 - **lsblk -f** → Filesystem info
-

7. Managing Swap

- Create swap file:

```
sudo fallocate -l 2G /swapfile
sudo chmod 600 /swapfile
sudo mkswap /swapfile
sudo swapon /swapfile
```

- Make it permanent in `/etc/fstab`:

```
/swapfile none swap sw 0 0
```

8. Logical Volume Management (LVM)

LVM allows dynamic disk management (create, resize volumes):

1. Create Physical Volume (PV)

```
sudo pvcreate /dev/sdb1
```

2. Create Volume Group (VG)

```
sudo vgcreate vgdata /dev/sdb1
```

3. Create Logical Volume (LV)

```
sudo lvcreate -L 10G -n lvdata vgdata
```

4. Format and mount:

```
sudo mkfs.ext4 /dev/vgdata/lvdata
sudo mount /dev/vgdata/lvdata /mnt
```

9. Graphical Tools

If using GUI:

- **GParted** – Partition editor

```
sudo apt install gparted # Debian/Ubuntu
```

- **Disks (gnome-disk-utility)** – Easy disk management
-

☒ Summary of Common Commands:

Task	Command
List disks	<code>lsblk, fdisk -l, parted -l</code>

Task	Command
Partition	<code>fdisk /dev/sdX, parted /dev/sdX</code>
Format	<code>mkfs.ext4 /dev/sdX1</code>
Mount	<code>mount /dev/sdX1 /mnt</code>
Check usage	<code>df -h, du -sh</code>
LVM	<code>pvcreate, vgcreate, lvcreate</code>

What is LVM?

LVM (Logical Volume Management) is a system for managing disk storage more flexibly than traditional partitioning.

Key features:

- Resize volumes dynamically (increase or decrease size)
- Combine multiple physical disks into a single storage pool
- Create snapshots for backups
- Simplifies disk management in large systems

2. LVM Components

Component	Description
Physical Volume (PV)	Actual physical storage device, like <code>/dev/sdb</code> or <code>/dev/sdc1</code>
Volume Group (VG)	Pool of storage created by combining one or more PVs
Logical Volume (LV)	Usable “virtual partition” created from a VG, can be formatted with a filesystem

Flow:

Physical Disk → PV → VG → LV → Filesystem → Mount

3. LVM Setup Steps

Step 1: Create Physical Volume (PV)

```
sudo pvcreate /dev/sdb
sudo pvdisplay      # Check PV info
```

Step 2: Create Volume Group (VG)

```
sudo vgcreate vgdata /dev/sdb
sudo vgdisplay      # Check VG info
```

Step 3: Create Logical Volume (LV)

```
sudo lvcreate -L 10G -n lvdata vgdata
sudo lvdisplay      # Check LV info
```

- -L 10G → size of LV
- -n lvdata → name of LV
- vgdata → VG from which LV is created

Step 4: Format Logical Volume

```
sudo mkfs.ext4 /dev/vgdata/lvdata
```

Step 5: Mount Logical Volume

```
sudo mkdir /mnt/data
sudo mount /dev/vgdata/lvdata /mnt/data
```

- To mount automatically at boot, edit /etc/fstab:

```
/dev/vgdata/lvdata /mnt/data ext4 defaults 0 2
```

4. Common LVM Commands

Task	Command
List PVs	<code>pvdisplay, pvs</code>
List VGs	<code>vgdisplay, vgs</code>
List LVs	<code>lvdisplay, lvs</code>
Extend LV	<code>lvextend -L +5G /dev/vgdata/lvdata</code>
Reduce LV	<code>lvreduce -L 5G /dev/vgdata/lvdata (careful!)</code>
Resize filesystem	<code>resize2fs /dev/vgdata/lvdata (after lvextend)</code>
Remove LV	<code>lvremove /dev/vgdata/lvdata</code>
Remove VG	<code>vgremove vgdata</code>
Remove PV	<code>pvremove /dev/sdb</code>

5. Resizing Logical Volumes

- **Increase LV size**

```
sudo lvextend -L +5G /dev/vgdata/lvdata
sudo resize2fs /dev/vgdata/lvdata
```

- **Decrease LV size** (dangerous, must shrink filesystem first)

```
sudo umount /mnt/data
sudo e2fsck -f /dev/vgdata/lvdata
sudo resize2fs /dev/vgdata/lvdata 5G
sudo lvreduce -L 5G /dev/vgdata/lvdata
sudo mount /dev/vgdata/lvdata /mnt/data
```

6. Snapshots

LVM allows creating **snapshots** for backups:

```
sudo lvcreate -L 2G -s -n lvdata_snap /dev/vgdata/lvdata
```

- -s → snapshot
 - Can mount snapshot to backup files safely
-

☒ Summary Diagram:

```
[Physical Disk /dev/sdb, /dev/sdc]
  ↓
Physical Volume (PV)
  ↓
Volume Group (VG) -> Pool of storage
  ↓
Logical Volume (LV) -> Like partitions
  ↓
Filesystem (ext4, xfs)
  ↓
Mounted at /mnt/data
```

Check Current Network Configuration

View IP addresses

```
ip addr show      # Shows all interfaces and their IPs
ifconfig          # Older command, may require net-tools
```

View routing table

```
ip route show
route -n
```

Check DNS settings

```
cat /etc/resolv.conf
```

Check network interfaces status

```
nmcli device status
```

2. Temporary Network Configuration

Temporary settings are lost after reboot.

Assign IP address

```
sudo ip addr add 192.168.1.10/24 dev eth0
```

Set default gateway

```
sudo ip route add default via 192.168.1.1
```

Bring interface up/down

```
sudo ip link set eth0 up
sudo ip link set eth0 down
```

3. Permanent Network Configuration

Permanent configuration depends on **Linux distribution**.

3.1 On Ubuntu/Debian (using netplan)

1. Config file: /etc/netplan/*.yaml

```
network:
  version: 2
  renderer: networkd
  ethernets:
```

```
eth0:
  dhcp4: no
  addresses: [192.168.1.10/24]
  gateway4: 192.168.1.1
  nameservers:
    addresses: [8.8.8.8, 8.8.4.4]
```

2. Apply configuration:

```
sudo netplan apply
```

3.2 On RHEL/CentOS/Fedora (using NetworkManager)

- Config files: `/etc/sysconfig/network-scripts/ifcfg-eth0`

Example:

```
DEVICE=eth0
BOOTPROTO=static
ONBOOT=yes
IPADDR=192.168.1.10
NETMASK=255.255.255.0
GATEWAY=192.168.1.1
DNS1=8.8.8.8
DNS2=8.8.4.4
```

- Restart network:

```
sudo systemctl restart network
```

4. DNS Configuration

- Edit `/etc/resolv.conf` (temporary, overwritten by NetworkManager)

```
nameserver 8.8.8.8
nameserver 8.8.4.4
```

- For permanent DNS, use `netplan` (Ubuntu) or `ifcfg` files (RHEL).
-

5. Test Network

Ping another device

```
ping 8.8.8.8          # Test connectivity to Internet
ping 192.168.1.1      # Test connectivity to gateway
```

Check open ports

```
netstat -tuln
```

```
ss -tuln
```

Check DNS resolution

```
nslookup google.com
dig google.com
```

6. Network Services Management

- Restart network services:

```
sudo systemctl restart NetworkManager # Ubuntu/CentOS
sudo systemctl restart networking      # Debian/Ubuntu older
```

- Enable/disable network interfaces permanently:

```
sudo nmcli device disconnect eth0
sudo nmcli device connect eth0
```

7. Wireless Network Configuration

- Scan Wi-Fi networks:

```
nmcli dev wifi list
```

- Connect to Wi-Fi:

```
nmcli dev wifi connect "SSID_NAME" password "PASSWORD"
```

☒ Summary Table:

Task	Temporary	Permanent
Assign IP	<code>ip addr add</code>	<code>netplan / ifcfg file</code>
Default gateway	<code>ip route add</code>	<code>netplan / ifcfg file</code>
DNS	<code>echo "nameserver 8.8.8.8"</code>	<code>netplan / ifcfg file</code>
Bring interface up/down	<code>ip link set eth0 up/down</code>	<code>ONBOOT=yes</code>
Check IP	<code>ip addr show</code>	same
Test connectivity	<code>ping, nslookup, dig</code>	same

ICNT ACADEMY PVT.LTD

What is SSH and Telnet?

Feature	SSH	Telnet
Full form	Secure Shell	Teletype Network
Security	Encrypted	Not encrypted (plain text)
Port	22	23
Authentication	Password / Key-based	Password only
Use	Secure remote login and command execution	Remote login (legacy, insecure)
Protocol	TCP	TCP

Key point: SSH is secure; Telnet should only be used on trusted networks for legacy devices.

2. SSH in Linux

2.1 Installing SSH Server

- **Ubuntu/Debian**

```
sudo apt update
sudo apt install openssh-server
```

- **RHEL/CentOS/Fedora**

```
sudo yum install openssh-server
sudo systemctl enable sshd
sudo systemctl start sshd
```

2.2 Check SSH Status

```
sudo systemctl status sshd
```

2.3 Connecting to Remote System

```
ssh username@remote_ip
```

Example:

```
ssh abhishek@192.168.1.10
```

2.4 SSH Key-based Authentication

1. Generate key pair:

```
ssh-keygen -t rsa -b 4096
```

2. Copy public key to remote server:

```
ssh-copy-id username@remote_ip
```

3. Login without password:

```
ssh username@remote_ip
```

2.5 SSH Config File

- **File:** ~/.ssh/config

Example:

```
Host server1
  HostName 192.168.1.10
  User abhishek
  Port 22
```

Then you can just type:

```
ssh server1
```

3. Telnet in Linux

3.1 Installing Telnet

- **Ubuntu/Debian**

```
sudo apt install telnet
```

- **RHEL/CentOS**

```
sudo yum install telnet
```

3.2 Connect to Remote Device

```
telnet remote_ip port
```

Example:

```
telnet 192.168.1.10 23
```

Note: Telnet sends data in plain text, including passwords, so it's not recommended on public networks.

4. Key Differences Between SSH and Telnet

Feature	SSH	Telnet
Encryption	Yes	No
Security	Very secure	Very insecure
Port	22	23
Password Safety	Encrypted	Plain text
Modern Use	Remote Linux server login	Rarely, legacy devices

5. SSH Best Practices

- Always use key-based authentication.
 - Disable root login via SSH (/etc/ssh/sshd_config → PermitRootLogin no).
 - Change default port 22 to something else (optional).
 - Use `fail2ban` to block repeated failed attempts.
-

6. Useful Commands

Task	Command
Check if SSH server is running	<code>systemctl status sshd</code>
Start/Restart SSH server	<code>systemctl start</code>
Connect via SSH	<code>ssh user@ip</code>
Copy files via SSH	<code>scp file user@ip:/path</code>
Connect via Telnet	<code>telnet ip port</code>
Check open port	<code>telnet ip port</code> or <code>nc -zv ip port</code>

. What is a Firewall?

A **firewall** is a security system that **controls incoming and outgoing network traffic** based on a set of rules.

- Protects servers from unauthorized access.
- Blocks unwanted connections while allowing legitimate traffic.

In Linux, common firewall tools are:

- **iptables** (legacy, very powerful)
- **firewalld** (RHEL/CentOS modern service)
- **ufw** (Ubuntu/Debian user-friendly frontend)

2. Checking Firewall Status

firewalld (RHEL/CentOS/Fedora)

```
sudo systemctl status firewalld
sudo firewall-cmd --state
```

ufw (Ubuntu/Debian)

```
sudo ufw status
```

3. Basic Firewall Concepts

Concept	Description
Zone	Grouping of interfaces with different trust levels (firewalld)
Port	Network port to allow/block traffic (e.g., 22 for SSH)
Service	Predefined ports/protocols like ssh, http, https
Rule	Instruction to allow/deny traffic

4. Configuring Firewall with firewalld

Start and Enable Firewall

```
sudo systemctl start firewalld
sudo systemctl enable firewalld
```

Check Default Zone

```
sudo firewall-cmd --get-default-zone
```

List Rules

```
sudo firewall-cmd --list-all
```

Open Port

```
sudo firewall-cmd --zone=public --add-port=80/tcp
sudo firewall-cmd --zone=public --add-port=80/tcp --permanent # persistent
sudo firewall-cmd --reload
```

Allow Services

```
sudo firewall-cmd --add-service=http --permanent
sudo firewall-cmd --add-service=https --permanent
sudo firewall-cmd --reload
```

Block Port

```
sudo firewall-cmd --remove-port=23/tcp --permanent
sudo firewall-cmd --reload
```

5. Configuring Firewall with ufw (Ubuntu/Debian)

Enable UFW

```
sudo ufw enable
```

Allow Ports/Services

```
sudo ufw allow 22 # SSH
sudo ufw allow 80 # HTTP
sudo ufw allow 443 # HTTPS
sudo ufw allow from 192.168.1.0/24 # Allow specific network
```

Deny Ports

```
sudo ufw deny 23 # Telnet
```

Check Status

```
sudo ufw status verbose
```

Disable UFW

```
sudo ufw disable
```

6. Configuring Firewall with iptables (Advanced)

Check Current Rules

```
sudo iptables -L -v
```

Allow Port

```
sudo iptables -A INPUT -p tcp --dport 22 -j ACCEPT
```

Block Port

```
sudo iptables -A INPUT -p tcp --dport 23 -j DROP
```

Save iptables Rules

```
sudo iptables-save > /etc/iptables/rules.v4
```

7. Testing Firewall

- **Ping Test:** Check if ICMP is allowed

```
ping google.com
```

- **Port Test:** Check if a port is open

```
telnet server_ip 22  
nc -zv server_ip 80
```

8. Summary Table

Tool	Commands	Notes
firewalld	firewall-cmd --add-port, --add-service	RHEL/CentOS/Fedora
ufw	ufw allow, ufw deny	Ubuntu/Debian simple
iptables	iptables -A INPUT ...	Advanced, very flexible

☒ Key Points:

- Firewalls filter traffic by **ports, protocols, IPs, or services**.
- **firewalld** uses zones, **ufw** is user-friendly, **iptables** is low-level.
- Always **allow SSH before enabling firewall** to avoid locking yourself out.

What is Package Management?

Package management is the process of **installing, updating, and removing software** on a Linux system. RHEL uses **RPM packages** and **YUM/DNF package managers**.

2. Package Types in RHEL

Package Type	Description
RPM (.rpm)	Red Hat Package Manager file, contains software binaries
Source (.src.rpm)	Source code RPM, used to build software
Repository (repo)	A collection of packages hosted online

3. RPM – Red Hat Package Manager

RPM is the low-level tool to **manage .rpm packages**.

Install an RPM

```
sudo rpm -ivh package.rpm
```

- **-i** → install
- **-v** → verbose
- **-h** → hash progress

Upgrade an RPM

```
sudo rpm -Uvh package.rpm
```

Remove a Package

```
sudo rpm -e package_name
```

Query Installed Packages

```
rpm -qa           # List all installed packages
rpm -qi package_name # Show package info
rpm -ql package_name # List files installed by package
```

Note: RPM does **not resolve dependencies automatically**.

4. YUM – Yellowdog Updater Modified

YUM is a higher-level package manager that **handles dependencies automatically**.

Basic YUM Commands

- **Install package**

```
sudo yum install package_name
```

- **Update package**

```
sudo yum update package_name
```

- **Update all packages**

```
sudo yum update
```

- **Remove package**

```
sudo yum remove package_name
```

- **Search for package**

```
yum search package_name
```

- **List installed packages**

```
yum list installed
```

- **List available packages**

```
yum list available
```

5. DNF – Next-Generation Package Manager (RHEL 8+)

DNF is the modern replacement for YUM, faster and more efficient.

Basic DNF Commands

- **Install:**

```
sudo dnf install package_name
```

- **Update:**

```
sudo dnf update package_name
```

- **Remove:**

```
sudo dnf remove package_name
```

- Search:

```
dnf search package_name
```

- List installed:

```
dnf list installed
```

- List available:

```
dnf list available
```

Note: Most YUM commands work with DNF as well.

6. Managing Repositories

Repositories are sources for packages.

List enabled repos

```
yum repolist  
dnf repolist
```

Add a repo

Create a `.repo` file in `/etc/yum.repos.d/`:

```
[myrepo]  
name=My Custom Repo  
baseurl=http://myrepo.example.com/repo/  
enabled=1  
gpgcheck=0
```

Then update cache:

```
sudo yum makecache  
sudo dnf makecache
```

7. Checking and Cleaning Cache

- Clean cached packages:

```
yum clean all  
dnf clean all
```

- Check package info:

```
yum info package_name  
dnf info package_name
```

8. Summary Table

Tool	Description	Key Commands
RPM	Low-level package manager	<code>rpm -ivh</code> , <code>rpm -Uvh</code> , <code>rpm -e</code> , <code>rpm -qa</code>
YUM	Handles dependencies	<code>yum install</code> , <code>yum update</code> , <code>yum remove</code> , <code>yum search</code>
DNF	Modern replacement for YUM	<code>dnf install</code> , <code>dnf update</code> , <code>dnf remove</code> , <code>dnf search</code>

☒ Key Points:

- **RPM** = low-level, manual dependency management.
- **YUM/DNF** = high-level, automatic dependency management.
- Repositories must be configured for YUM/DNF to fetch packages.
- RHEL 8+ prefers **DNF**, older versions use **YUM**.

Prerequisites

- A RHEL/CentOS server (YUM server) with a static IP, e.g., 192.168.1.100
 - Enough storage to store RPM packages
 - Root or sudo privileges
-

2. Install Required Packages

Install `createrepo` to create the YUM repository:

```
sudo yum install -y createrepo httpd
```

- `createrepo` → creates repository metadata
 - `httpd` → optional, if you want to serve repo via HTTP
-

3. Create Directory for Repository

```
sudo mkdir -p /var/www/html/yumrepo
```

- `/var/www/html/yumrepo` → folder to store RPM packages
-

4. Copy RPM Packages to Repository

You can either:

- Copy downloaded RPM packages from ISO or other sources:

```
sudo cp /path/to/rpms/*.rpm /var/www/html/yumrepo/
```

- Or mount RHEL ISO and copy all packages:

```
sudo mount -o loop /path/to/rhel.iso /mnt
sudo cp -r /mnt/Packages/* /var/www/html/yumrepo/
```

5. Create Repository Metadata

```
sudo createrepo /var/www/html/yumrepo/
```

- This generates `repodata/` folder inside `yumrepo/`
-

6. Configure HTTP Server (Optional)

If you want clients to access repo via HTTP:

- Start and enable `httpd`:

```
sudo systemctl start httpd
sudo systemctl enable httpd
```

- Adjust firewall to allow HTTP:

```
sudo firewall-cmd --permanent --add-service=http
sudo firewall-cmd --reload
```

- Verify repository is accessible:

Open in browser or use `curl`:

```
curl http://192.168.1.100/yumrepo/
```

7. Create Repo File on Client

On client machines, create a `.repo` file in `/etc/yum.repos.d/`:

```
sudo nano /etc/yum.repos.d/local.repo
```

Add content:

```
[localrepo]
name=Local YUM Repository
baseurl=http://192.168.1.100/yumrepo/
enabled=1
gpgcheck=0
```

- `baseurl` → your server IP + repository path
 - `gpgcheck=0` → disable GPG verification (for testing, optional)
-

8. Test Repository on Client

```
sudo yum clean all
sudo yum repolist
sudo yum install package_name
```

- `yum repolist` should show `localrepo`
 - You can now install RPMs from your local YUM server
-

9. Optional: Automate Repository Updates

If you add new RPMs to `/var/www/html/yumrepo/`, update metadata:

```
sudo createrepo --update /var/www/html/yumrepo/
```

Summary Diagram (Flow)

```
RPM Packages (Server)
  ↓
createrepo -> /var/www/html/yumrepo
  ↓
HTTP Server (http://192.168.1.100/yumrepo/)
  ↓
Client yum repo (/etc/yum.repos.d/local.repo)
  ↓
yum install package_name
```

☑ Key Points:

- YUM server uses `createrepo` to generate metadata.
- HTTP is the easiest way for clients to access the repo.
- Firewall must allow HTTP (port 80).
- Clients just need `.repo` file pointing to server IP.

What is NFS?

NFS (Network File System) allows a **Linux/Unix system to share directories and files over a network.**

Key points:

- Developed by Sun Microsystems.
- Works on client-server architecture.
- Allows multiple clients to access the same files as if they were local.

2. NFS Components

Component	Description
NFS Server	The system that shares directories
NFS Client	The system that mounts shared directories
Export	Directory on server made available to clients
Mount	Accessing the exported directory on client

3. Install NFS Packages

RHEL/CentOS

```
sudo yum install -y nfs-utils
```

Ubuntu/Debian

```
sudo apt update
sudo apt install -y nfs-kernel-server
```

4. Configure NFS Server

Step 1: Create Directory to Share

```
sudo mkdir -p /srv/nfs/shared
sudo chown nobody:nogroup /srv/nfs/shared
sudo chmod 777 /srv/nfs/shared
```

Step 2: Configure Exports

Edit /etc/exports file:

```
/srv/nfs/shared 192.168.1.0/24(rw,sync,no_root_squash)
```

- /srv/nfs/shared → directory to share
- 192.168.1.0/24 → allowed client network
- rw → read/write access
- sync → writes are synchronous
- no_root_squash → root on client can access as root

Step 3: Export Directories

```
sudo exportfs -a
```

Step 4: Start NFS Server

```
sudo systemctl enable nfs-server
sudo systemctl start nfs-server
sudo systemctl status nfs-server
```

5. Configure Firewall on Server

Allow NFS ports:

```
sudo firewall-cmd --permanent --add-service=nfs
sudo firewall-cmd --permanent --add-service=mountd
sudo firewall-cmd --permanent --add-service=rpc-bind
sudo firewall-cmd --reload
```

6. Configure NFS Client

Step 1: Install NFS Client

- **RHEL/CentOS**

```
sudo yum install -y nfs-utils
```

- **Ubuntu/Debian**

```
sudo apt install -y nfs-common
```

Step 2: Mount NFS Share

```
sudo mkdir -p /mnt/nfs
sudo mount 192.168.1.100:/srv/nfs/shared /mnt/nfs
```

- 192.168.1.100 → NFS server IP
- /mnt/nfs → local mount point

Step 3: Verify Mount

```
df -h
mount | grep nfs
```

Step 4: Permanent Mount

Add entry to `/etc/fstab`:

```
192.168.1.100:/srv/nfs/shared /mnt/nfs nfs defaults 0 0
```

7. Useful NFS Commands

Task	Command
Show exported directories	<code>showmount -e 192.168.1.100</code>
Mount NFS share	<code>mount server_ip:/export /mnt</code>
Unmount NFS share	<code>umount /mnt/nfs</code>
Check NFS mounts	<code>df -hT</code>
Re-export after changes	<code>exportfs -ra</code>

8. NFS Tips

- Always check client IP/network in `/etc/exports`.
- Use `sync` option for data integrity.
- `no_root_squash` should be used carefully; default is `root_squash`.
- NFS works best in trusted internal networks.

9. Summary Diagram (Flow)

```
[Server] 192.168.1.100
  /srv/nfs/shared ---> Exported via NFS
    |
    v
[Client] 192.168.1.101
  /mnt/nfs ---> Mounted directory
```

- Clients access files as if they are local.
- Multiple clients can share the same NFS directory.

What is DHCP?

DHCP (Dynamic Host Configuration Protocol) is a network protocol that **automatically assigns IP addresses and other network settings** (like gateway, DNS) to clients in a network.

Key Points:

- Reduces manual IP configuration.
 - Prevents IP conflicts.
 - Works in client-server architecture.
-

2. Install DHCP Server

RHEL/CentOS

```
sudo yum install -y dhcp
```

Ubuntu/Debian

```
sudo apt update  
sudo apt install -y isc-dhcp-server
```

3. Configure DHCP Server

Step 1: Edit DHCP Configuration File

- **RHEL/CentOS:** /etc/dhcp/dhcpd.conf
- **Ubuntu/Debian:** /etc/dhcp/dhcpd.conf

Example configuration:

```
# DHCP default settings  
option domain-name "example.com";  
option domain-name-servers 8.8.8.8, 8.8.4.4;  
  
default-lease-time 600;           # 10 minutes  
max-lease-time 7200;             # 2 hours  
  
# Network/subnet declaration  
subnet 192.168.1.0 netmask 255.255.255.0 {  
    range 192.168.1.100 192.168.1.200;  
    option routers 192.168.1.1;  
    option broadcast-address 192.168.1.255;  
}
```

Explanation:

- range → IP addresses to assign dynamically

- option routers → default gateway
 - option domain-name-servers → DNS servers
-

Step 2: Configure DHCP Server to Listen on Interface

- **RHEL/CentOS:** Edit `/etc/sysconfig/dhcpd`

```
DHCPDARGS=eth0
```

- **Ubuntu/Debian:** Edit `/etc/default/isc-dhcp-server`

```
INTERFACESv4="eth0"
```

Replace `eth0` with your server network interface.

Step 3: Start and Enable DHCP Service

```
sudo systemctl start dhcpd          # RHEL/CentOS
sudo systemctl enable dhcpd
```

```
sudo systemctl start isc-dhcp-server # Ubuntu/Debian
sudo systemctl enable isc-dhcp-server
```

4. Configure Firewall

Allow DHCP traffic (UDP ports 67 and 68):

```
sudo firewall-cmd --add-service=dhcp --permanent
sudo firewall-cmd --reload
```

5. Verify DHCP Server

- **Check service status**

```
sudo systemctl status dhcpd
```

- **Check assigned leases**

```
cat /var/lib/dhcp/dhcpd.leases
```

- **Check network interface on client**

```
ip addr show
# or use dhclient to request IP
sudo dhclient -v
```

6. Assign Static IP to a Client (Optional)

Add a host section in `/etc/dhcp/dhcpd.conf`:

```
host client1 {  
    hardware ethernet 00:11:22:33:44:55;  
    fixed-address 192.168.1.50;  
}
```

- hardware ethernet → MAC address of client
 - fixed-address → IP to assign
-

7. Useful Commands

Task	Command
Start service	<code>systemctl start dhcpd</code>
Enable service	<code>systemctl enable dhcpd</code>
Check status	<code>systemctl status dhcpd</code>
View leases	<code>cat /var/lib/dhcp/dhcpd.leases</code>
Request IP on client	<code>dhclient -v eth0</code>
Restart service	<code>systemctl restart dhcpd</code>

8. DHCP Server Workflow

```
[Client] ---- DHCPDISCOVER ---> [Server]  
[Server] ---- DHCPOFFER ---> [Client]  
[Client] ---- DHCPREQUEST ---> [Server]  
[Server] ---- DHCPACK ---> [Client]
```

- Client requests IP → Server offers IP → Client accepts → Server acknowledges
-

☒ Key Points:

- DHCP avoids manual IP assignment.
- DHCP leases can be dynamic or static.
- Proper firewall configuration is necessary.
- Verify client IP to ensure DHCP works.

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What is Samba?

Samba is a software that allows **Linux/Unix systems to share files and printers with Windows systems** using the **SMB/CIFS protocol**.

Key points:

- Enables cross-platform file sharing.
 - Can be used for home networks or enterprise environments.
 - Works as **server** and allows **user authentication**.
-

2. Install Samba

RHEL/CentOS

```
sudo yum install -y samba samba-client samba-common
```

Ubuntu/Debian

```
sudo apt update  
sudo apt install -y samba
```

3. Create a Directory to Share

```
sudo mkdir -p /srv/samba/shared  
sudo chown nobody:nogroup /srv/samba/shared  
sudo chmod 2770 /srv/samba/shared
```

- `chmod 2770` → allows group read/write and sets group ID
 - `chown nobody:nogroup` → default guest user (optional)
-

4. Configure Samba

Edit the Samba configuration file:
`/etc/samba/smb.conf`

Add at the end:

```
[SharedFiles]  
path = /srv/samba/shared  
browseable = yes  
read only = no  
guest ok = yes
```

Explanation:

- `[SharedFiles]` → share name visible to clients

- path → folder path
 - browseable=yes → visible in network
 - read only=no → allow write access
 - guest ok=yes → no authentication (optional)
-

5. Configure Samba User (Optional)

Samba can have **user-level authentication**:

1. Add Linux user:

```
sudo useradd sambauser  
sudo passwd sambauser
```

2. Add Samba user:

```
sudo smbpasswd -a sambauser  
sudo smbpasswd -e sambauser
```

3. Update directory permissions:

```
sudo chown sambauser:sambauser /srv/samba/shared
```

6. Start and Enable Samba Services

```
sudo systemctl start smb nmb  
sudo systemctl enable smb nmb  
sudo systemctl status smb nmb
```

- smb → handles file sharing
 - nmb → handles NetBIOS name resolution
-

7. Configure Firewall

Allow Samba traffic:

```
sudo firewall-cmd --permanent --add-service=samba  
sudo firewall-cmd --reload
```

- Ports opened: TCP 139, TCP 445, UDP 137, UDP 138
-

8. Access Samba Share from Client

Linux Client

```
sudo mount -t cifs //192.168.1.100/SharedFiles /mnt/samba -o  
username=sambauser,password=yourpassword
```

Windows Client

1. Open Run → \\192.168.1.100\SharedFiles
2. Enter Samba username/password if prompted

9. Useful Commands

Task	Command
Test configuration	testparm
List active shares	smbstatus
Restart Samba	systemctl restart smb nmb
Check firewall	firewall-cmd --list-all
Add Samba user	smbpasswd -a username

10. Summary Diagram (Flow)

```
[Linux Samba Server]  
  /srv/samba/shared  ---> Shared via SMB  
      |  
      v  
[Clients]  
  Linux: mount -t cifs  
  Windows: \\server_ip\SharedFiles
```

☒ Key Points:

- Samba allows file sharing across Linux and Windows.
- Can be **guest access** or **authenticated user access**.
- Proper firewall configuration is necessary.
- Use testparm to verify smb.conf syntax before restarting services.